

# JHASHANK NAYAN

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## Summary

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Computer Science (Data Science) undergraduate with hands-on experience in software development, backend engineering, REST API development, and AI/ML systems. Proficient in Python, Java, SQL, FastAPI, React, PyTorch, and computer vision, with experience building inference pipelines, integrating software components, optimizing system workflows, and evaluating model performance.

## Education

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**Acharya Institute of Technology**, Bangalore  
Bachelor of Engineering in Computer Science (Data Science)

Sept. 2023 – Present  
**CGPA: 9.2/10**

## Skills

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**Programming:** Python, Java, SQL

**Software Engineering:** Data Structures & Algorithms, Object-Oriented Programming, REST APIs, Backend Development, API Integration

**Backend:** FastAPI, Flask, Node.js, Uvicorn

**Frontend & Mobile:** React.js, Android, Kotlin, Vite

**Machine Learning:** PyTorch, Scikit-learn, OpenCV, MediaPipe, MoveNet, TensorFlow Lite, CNNs, Vision Transformers, OCR

**AI & Information Retrieval:** RAG, Vector Search, BM25, Cross-Encoder Reranking, Transformers, Local LLMs, Ollama

**Tools:** Git, GitHub, Linux, pytest, NumPy, SciPy, PennyLane

## Projects

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**Deepfake Detection Pipeline for KYC: rPPG + Hybrid Quantum-Classical ML** [github.com/NYN-05/Major\\_project](https://github.com/NYN-05/Major_project)

*Python, PyTorch, PennyLane, OpenCV, MediaPipe, React, scikit-learn*

- Built an end-to-end low-resolution KYC video detection pipeline integrating frame processing, rPPG-based physiological analysis, and hybrid quantum-classical ML inference.
- Engineered POS/CHROM signal processing and extracted **23 physiological features** for temporal liveness and deepfake analysis.
- Improved the VQC from majority-class prediction to genuine FAKE-class detection, achieving **0.702 specificity**; integrated inference with a React-based application.

**VeriSight V1 – AI-Powered Image Authenticity Platform** [github.com/NYN-05/VERISIGHT\\_V1](https://github.com/NYN-05/VERISIGHT_V1)

*Python, FastAPI, PyTorch, OpenCV, React, CNN, ViT, OCR*

- Developed an image-verification platform combining CNN, Vision Transformer, and OCR models through a **FastAPI REST backend**.
- Implemented asynchronous and parallel model inference with caching, concurrency controls, benchmark evaluation, latency measurement, and threshold calibration.

**Preventive Movement Intelligence – BIRAC Idea Prototype** [github.com/NYN-05/BIRAC\\_IDEA\\_COMPLETE](https://github.com/NYN-05/BIRAC_IDEA_COMPLETE)

*Python, FastAPI, MoveNet, MediaPipe, TensorFlow Lite, Kotlin, Android*

- Developed an Android-to-FastAPI real-time posture analysis system using MoveNet and MediaPipe for pose inference, movement scoring, and risk analysis.
- Implemented exercise-aware assessment for squats, push-ups, and sit-ups with form scoring, movement-stage detection, risk estimation, and corrective feedback.

## Achievements

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- **2nd Place, AICTE Ideathon 2025** – team project on a community cloud-kitchen resource-sharing platform, evaluated at the college level.
- **Certificate of Appreciation for Outstanding Academic Performance, 2024.**